TÉCNICO LISBOA

Mathematics for Machine Learning

2025-2026

Project 2 - A deeper understanding

In this course you learned and are still learning something (we hope) about several mathematical frameworks with fundamental application to machine learning. The subject matter we covered is never-the-less just a slice of the most fundamental ideas. Due to time constraints, there are many beautiful and useful mathematical theories that we did not cover, and there are also deeper results or applications in the subject matter we did cover.

In this project you have the opportunity to choose a theme you are interested in for a deeper look and understanding. Below are several suggestions, but you might also suggest a theme of your own. For this project you can use Python or R, as suits the theme better.

You have to submit by the deadline of **June 2 (1 pm)** a zip file containing:

- An article-type 8 page report (graphics included) that describes your work, including methodology, analysis, and discussion of your results;
- The commented code of your project;
- A 10 minute presentation to be then delivered in class on **June 2** (starting at **2 pm**).

This assignment is **by Groups of 2**. A maximum of three groups can choose the same topic, first came, first served. Choose your topic and enroll your group until **May 4** (midnight- 1 minute) in the following document:
https://docs.google.com/spreadsheets/d/11z0y92nnPwu_e4SDkGUJUYZGYgiCYBQsHmg69a3dDoY/edit?usp=sharing

List of topics

1 - Multivariate outlier explanations using Shapley values and Mahalanobis distances

Marcus Mayrhofer, Peter Filzmoser,

Multivariate outlier explanations using Shapley values and Mahalanobis distances, Econometrics and Statistics, 2023,
https://doi.org/10.1016/j.ecosta.2023.04.003.
Link: https://www.sciencedirect.com/science/article/pii/S2452306223000291

Tasks:

- **Read the paper**: Carefully read the paper to understand how Shapley values are used to explain Mahalanobis distances and their role in identifying and interpreting multivariate outliers.
- **Replicate a part of the simulation study**: Select and reproduce a part of the simulation study, ensuring a comparison between classical and robust approaches in terms of their ability to detect and explain outliers.
- **Application to Real Data**: Select a dataset. Apply both classical and robust Mahalanobis distances to detect outliers. Compute the associated Shapley value and discuss its interpretations. Discuss the ability of each method to identify and interpret outliers.
- **Evaluation and Critical Discussion**: Compare the outcomes of robust vs. non-robust methods.Discuss the ability of each method to identify and interpret outliers. Do a critical assessment of the strengths and limitations of the methods under study.

2 - Robust Covariance Estimation and Explainable Outlier Detection for Matrix-Valued Data

Mayrhofer, M., Radojičić, U., Filzmoser, P. (2025). Robust Covariance Estimation and Explainable Outlier Detection for Matrix-Valued Data. *Technometrics*, 1–15. https://doi.org/10.1080/00401706.2025.2475781
Link: https://www.tandfonline.com/doi/full/10.1080/00401706.2025.2475781

Software and data availability: The R package *robustmatrix* includes a parallelized C++ implementation of the MMCD algorithm (Mayrhofer, Radojičić, and Filzmoser 2024). The vignette MMCD_examples and the R code in the supplementary material replicate the examples presented in this paper. Please, read also the supplementary material.

Mayrhofer, M., Radojičić, U., Filzmoser, P. (2024), robustmatrix: RobustMatrix-Variate Parameter Estimation R Package Version 0.1.3. Available at https://cran.r-project.org/web/packages/robustmatrix/index.html

Tasks:

- **Read the paper**: Carefully read the paper to understand how Matrix Minimum Covariance estimator was derived. Understand its use in outlier detection of matrix-valued data. Understand how Shapley values are derived and their role in identifying and interpreting multivariate outliers.
- **Replicate a part of the simulation study**: Select and reproduce a part of the simulation study, ensuring a comparison between classical and robust approaches in terms of their ability to detect and explain outliers.
- **Application to Real Data**: Select a dataset. Apply both classical and robust Mahalanobis distances to detect outliers. Compute the associated Shapley value and discuss its interpretations. Discuss the ability of each method to identify and interpret outliers.
- **Evaluation and Critical Discussion**: Compare the outcomes of robust vs. non-robust methods.Discuss the ability of each method to identify and interpret outliers. Do a critical assessment of the strengths and limitations of the methods under study.

3 - MacroPCA

Hubert, M., Rousseeuw, P.J., Van den Bossche, W. (2019). MacroPCA: An all-in-one PCA method allowing for missing values as well as cellwise and
rowwise outliers, Technometrics, 61, 459–473.
Link:
https://wis.kuleuven.be/stat/robust/papers/publications-2019/hubertetal-macropca-technometrics-open-access.pdf

See the slides available at:
https://sites.google.com/view/joclad2025-en/scientific-program/plenary-sessions
And
https://sites.google.com/view/joclad2025-en/mini-course

R package: The methods described in the previous article as well as the DPOSS data have been made available in the R package *cellWise* (Raymaekers et al. 2019) on CRAN, with a vignette reproducing the examples shown.

Raymaekers, J., & Rousseeuw, P. J. (2023b). cellWise: Analyzing Data with Cellwise Outliers [Computer software manual]. Retrieved from https://CRAN.R-project.org/package=cellWise

Tasks:

- **Read the paper**: Carefully read the paper to understand what is a casewise and a cellwise outliers as well as the robust principal component proposed method.

- **Replicate a part of the simulation study**: Select and reproduce a part of the simulation study, ensuring a comparison between classical and robust approaches in terms of their ability to detect and explain outliers.
- **Application to Real Data**: Select a dataset. Apply both classical and robust methods to detect outliers.
- Introduce 10% of outliers and missing observations into your dataset by inserting observations that deviate significantly from the typical structure of the data. Repeat the analysis with the contaminated dataset and evaluate the effectiveness of the robust estimates. Analyse the detected outliers and compare with the known ground truth. Discuss the ability of each method to identify and interpret outliers.
- **Evaluation and Critical Discussion**: Compare the outcomes of robust vs. non-robust methods. Discuss the ability of each method to identify and interpret outliers. Do a critical assessment of the strengths and limitations of the methods under study.

4 - Class maps for visualizing classification results

Raymaekers, J., Rousseeuw, P.J., Hubert, M. (2022). Class maps for visualizing classification results. Technometrics, 64, 151-165. https://www.tandfonline.com/doi/full/10.1080/00401706.2021.1927849

R package: The methods described in the previous article have been made available in the R package *classmap*. See also the Supplementary material of the paper.

Raymaekers, J., & Rousseeuw, P. J. (2023c). classmap: Visualizing Classification Results [Computer software manual]. Retrieved from https://CRAN.R-project.org/package=classmap

Tasks:

- **Read the Paper**: Carefully read the paper to understand the visualization tools proposed to contribute with additional interpretation of the confusion matrices in classification problems. Understand the concepts of “PAC” and “farness.”
- **Application to Real Data**: Select two datasets suitable for classification tasks and choose three classifiers discussed in the paper. Apply the proposed methodologies to both datasets and analyze the results. Introduce outliers into each dataset in two distinct ways: (i) by inserting an observation within a class that deviates significantly from the typical structure of that class, and (ii) by altering the class label of an observation (label switching). Repeat the analysis with the modified datasets and evaluate the effectiveness of the visualization tools in detecting and explaining these outliers. What happens if instead of one observation you apply the contamination scheme to 10% of the observations of the selected class?

- **Evaluation and Critical Discussion**: Compare the results across the different experiments. Discuss the extent to which the visualization tools help identify and interpret outliers. Provide a critical assessment of the strengths and limitations of the proposed methods.

5 - A robust regression method based on exponential-type kernel functions

Francisco de A.T. de Carvalho, Eufrásio de A. Lima Neto, and Marcelo R.P. Ferreira. A robust regression method based on exponential-type kernel functions, *Neurocomputing*, 234, 2017, https://doi.org/10.1016/j.neucom.2016.12.035

Link: **https://doi.org/10.1016/j.neucom.2016.12.035**

Software: Authors stated that the code is available on GitHub through the link https://github.com/marcelorpf/gkrreg.

Tasks:

- **Read the paper**: Carefully examine the paper to understand the use of exponential-type kernels in enhancing the robustness of linear regression models. Give special attention to the challenges involved in hyperparameter selection.
- **Replicate a part of the simulation study**: Select and reproduce a part of the simulation study, ensuring a comparison between classical and robust approaches in terms of their ability to cope with outliers.
- **Application to Real Data**: Select a dataset. Apply both classical, the proposed robust linear regression, and any of the other benchmarks used in the paper.
- Introduce 10% of outliers into your dataset by inserting observations that deviates significantly from the typical structure of the data. Repeat the analysis with the contaminated dataset and evaluate the effectiveness of the robust estimates. Analyse the detected outliers and compare with the known ground truth. Discuss the ability of each method to identify the outliers.
- **Evaluation and Critical Discussion**: Compare the outcomes of robust vs. non-robust methods. Discuss the ability of each method to identify and interpret outliers. Do a critical assessment of the strengths and limitations of the methods under study.

6 - Euclidian Distance Matrices: applications and completion problem

Andrés David Báez Sánchez, Carlile Lavor (2020) On the estimation of unknown distances for a class of Euclidean distance matrix completion problems with interval data. *Linear Algebra and its Applications*, Volume 592, Pages 287-305, ISSN 0024-3795, https://doi.org/10.1016/j.laa.2020.01.036.

Link: https://www.sciencedirect.com/science/article/pii/S002437952030046X?via%3Dihub

Further readings:

Dokmanic, I., Parhizkar, R., Ranieri, J., & Vetterli, M. (2015). Euclidean distance matrices: essential theory, algorithms, and applications. IEEE Signal Processing Magazine, 32(6), 12-30. https://arxiv.org/pdf/1502.07541.

Candes, E. J., & Recht, B. (2008). Exact low-rank matrix completion via convex optimization. In Allerton (pp. 806-812).https://link.springer.com/article/10.1007/s10208-009-9045-5.

Software: We suggest you write your code in R.

Tasks:

- **Read the paper**: Carefully read the paper to understand the proposed completion solution for the given class of EDM.
- **Replicate the examples 5 and 6**: Reproduce the simulation study, using a R package, and try to explain the chosen entries of Example 6.
- **Application to Real Model**: Check the main applications of EDMs, for instance, molecular conformation, sensor network localization or multidimensional scaling and use real data, if possible, to addapt to the proposed completion solution.
- **Evaluation and Critical Discussion**: Read about (an)other completion method(s) (see Other readings) Compare the outcomes for the model you picked in the previous step. Do a critical assessment of the strengths and limitations of the methods under consideration.

7- Mathematical Principles of Large Language Models (LLMs)
Vaswani et al. (2017). Attention is All You Need
https://arxiv.org/abs/1706.03762
Devlin et al. (2018). BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding
https://arxiv.org/abs/1810.04805
You should also watch: https://www.3blue1brown.com/lessons/attention
https://www.youtube.com/watch?v=PSs6nxngL6k

Tasks:

- Read the papers, watch the videos and summarize the key concepts introduced, including self-attention mechanism, positional encoding, and the overall architecture of the Transformer model. Explain how these concepts contribute to the functionality of LLMs
- Derive the mathematical formula for the scaled dot-product attention mechanism described in the paper. Show all steps clearly.
- Explain the role of the softmax function in the attention mechanism and its effect on the model's output.
- Construct a simple numerical example (you can use small, fabricated matrices) to demonstrate how the attention mechanism works in practice. Perform the calculations manually or using a simple Python script.
- Write a Python function using only numpy to simulate a single-headed attention mechanism, as is done in the second vídeo. Use a 3 dimensional embedding space and a very simple 1st Grade vocabulary.

Further Reading:
Radford et al. (2019). GPT-2: Language Models are Unsupervised Multitask Learners
https://cdn.openai.com/better-language-models/language_models_are_unsupervised_multitask_learners.pdf

8 (Challenge 1) & 9 (Challenge 2)- Automated Analogy Discovery and Relational Mapping
Sultan, O., & Shahaf, D. (2022). *Life is a Circus and We are the Clowns: Automatically Finding Analogies between Situations and Processes*. Proceedings of EMNLP. Link
Software: The authors provide the code and datasets on GitHub:
https://github.com/orensul/analogies_mining

Tasks

- **Read and Comprehend the Structural Mapping Theory (SMT):** Carefully examine the paper to understand how the authors move beyond simple word analogies (A:B :: C:D) to complex process analogies.

- **Focus Point:** Give special attention to the "Analogous Matching Algorithm" and how it uses a combination of semantic similarity (SBERT) and structural constraints to find an optimal mapping M.
- **Challenge 1:** Understand the challenge of "relational similarity" vs. "attribute similarity"—i.e., why "blood" should map to "water" in a heart/pump analogy even though they are semantically different substances.
- **Challenge 2:** Implement this framework (similarities of similarities) with a convolutional model. See, similar CNN models in
  -Lim, S., et al.: Classifying and completing word analogies by machine learning. IJAR 132, 1–25 (2021)
  https://www.sciencedirect.com/science/article/am/pii/S0888613X21000141
  -Esteban Marquer, et al.: *Solving morphological analogies: from retrieval to generation*. Ann. Math. Artif. Intell. 93(2): 263-298 (2025)
  https://hal.science/hal-04056908/file/AMAI_Special_issue_2023___Morphological_analogy_solving.pdf
- **Replicate the Mapping Evaluation:** Select a small subset of the ProPara dataset or the Cognitive Psychology Stories provided in the repository.
- **Task:** Reproduce the mapping process for a specific pair of processes (e.g., "How the immune system works" vs. "Photosynthesis").
- **Comparison:** Compare the algorithm's output against a baseline that only uses "Maximum Similarity" (mapping entities based solely on the highest SBERT score) to see how often surface-level similarity fails compared to the paper's relational approach.
- **Real world use-cases** Select two procedural texts from a domain not heavily featured in the paper, e.g., immunity and cybersecurity systems. Find analogies between them and discuss them.

10 - Open Topic (max. 2 groups)

- Requires 1 or 2 out-of-class meeting with the group
- Make sure to have an interesting work/papers and datasets to explore, a few interesting questions to pursue and a feasible goal in mind